Count Pairs With Sum Less Than Target

Java Code

```
import java.util.Arrays;

class Solution {
    int countPairs(int arr[], int target) {

        Arrays.sort(arr);

        int i = 0;
        int j = arr.length - 1;
        int count = 0;

        while (i < j) {

            int sum = arr[i] + arr[j];

            if (sum < target) {
                count += j - i;
                i++;
            }
            else {
                j--;
            }
        }

        return count;
    }
}
```

Visualization

Input

```
arr = [2, 1, 8, 3, 4, 7, 6, 5]
target = 7
```

Step 1: Sort

```
Before:
[2, 1, 8, 3, 4, 7, 6, 5]

After:
[1, 2, 3, 4, 5, 6, 7, 8]
 ↑                       ↑
 i                       j
```

Step 2

```
1 + 8 = 9

9 < 7 ❌

[1, 2, 3, 4, 5, 6, 7, 8]
 ↑                    ↑
 i                    j

j--
```

Step 3

```
1 + 7 = 8

8 < 7 ❌

[1, 2, 3, 4, 5, 6, 7, 8]
 ↑                 ↑
 i                 j

j--
```

Step 4

```
1 + 6 = 7

7 < 7 ❌
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
 ↑              ↑
 i              j

j--
```

Step 5 — Important

```
1 + 5 = 6

6 < 7 ✅
```

Since the array is sorted:

```
1 + 2 = 3 ✅
1 + 3 = 4 ✅
1 + 4 = 5 ✅
1 + 5 = 6 ✅
```

Visualization:

```
[1, 2, 3, 4, 5, 6, 7, 8]
 ↑  ↑  ↑  ↑  ↑
 i              j

Valid pairs:
   └──┴──┴──┴──┘

Number of pairs = j - i
                 = 4 - 0
                 = 4
```

```
count = 4
i++
```

Step 6

```
[1, 2, 3, 4, 5, 6, 7, 8]
    ↑           ↑
    i           j

2 + 5 = 7

7 < 7 ❌

j--
```

Step 7

```
[1, 2, 3, 4, 5, 6, 7, 8]
    ↑        ↑
    i        j

2 + 4 = 6

6 < 7 ✅
```

Valid pairs:

```
2 + 3 = 5 ✅
2 + 4 = 6 ✅
```

```
count += j - i
       = 3 - 1
       = 2
```

```
count = 4 + 2
      = 6
```

Final Visualization

```
Sorted Array:

[1, 2, 3, 4, 5, 6, 7, 8]
 ↑              ↑
 i              j

1 + 5 < 7
   ↓
[1,2] [1,3] [1,4] [1,5]
        4 pairs


[1, 2, 3, 4, 5, 6, 7, 8]
    ↑        ↑
    i        j

2 + 4 < 7
   ↓
[2,3] [2,4]
        2 pairs
```

Total

```
4 + 2 = 6
```

```
Output = 6
```

Pointer Movement

```
sum >= target
      ↓
     j--

sum < target
      ↓
count += j - i
```

```
    ↓
   i++
```

Complexity

```
Sorting       → O(n log n)
Two Pointers  → O(n)

Total         → O(n log n)
```